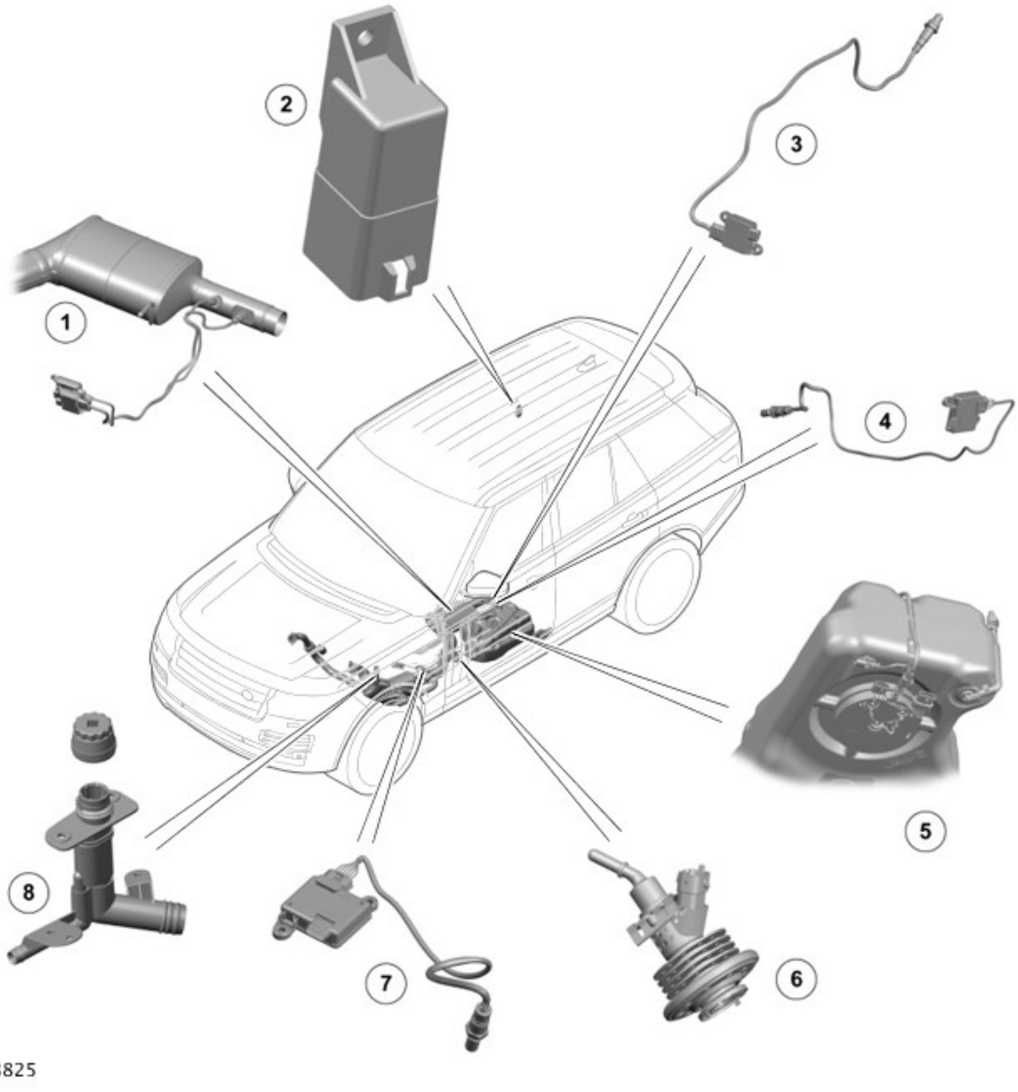
2016.0 RANGE ROVER (LG), 309-00

EXHAUST SYSTEM – TDV6 3.0L DIESEL

DESCRIPTION AND OPERATION

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Selective Catalytic Reduction (SCR) catalytic converter
2	Diesel Exhaust Fluid (DEF) heater control unit
3	Post - selective catalytic reduction (SCR) NOx sensor
4	Post - selective catalytic reduction (SCR) soot sensor (NAS market vehicles only)
5	DEF tank with DEF tank module
6	DEF injector
7	Pre - selective catalytic reduction NOx sensor (NAS market vehicles only)
8	DEF filler neck

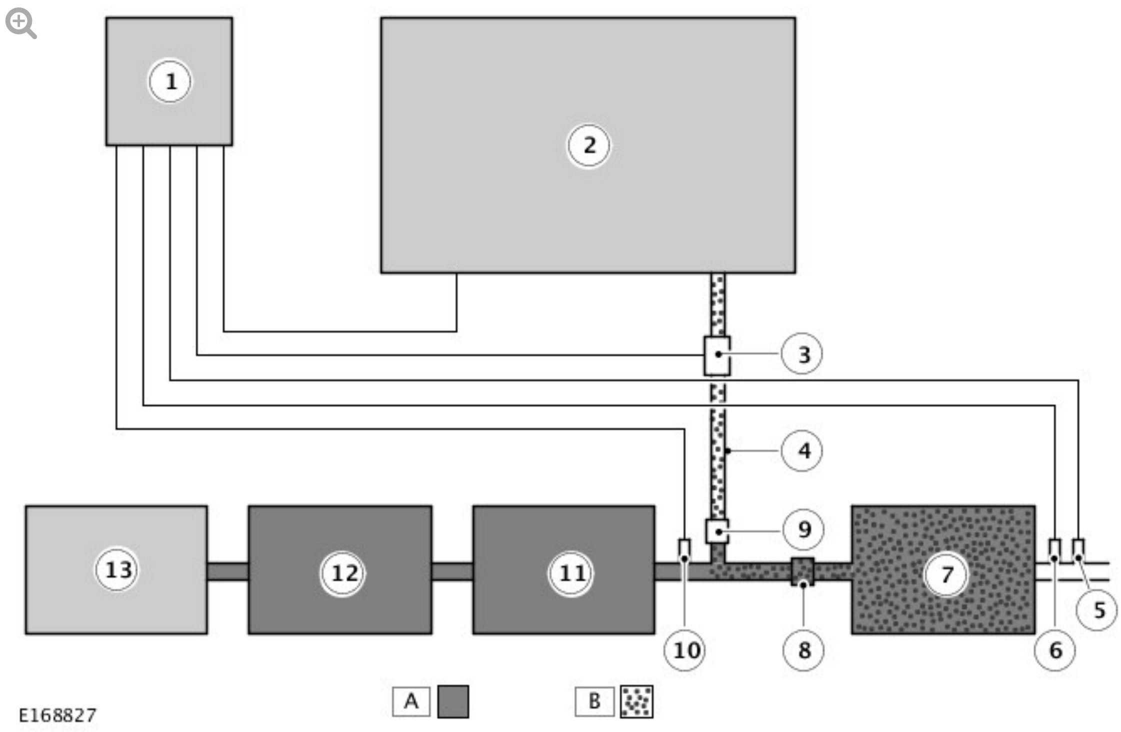
OVERVIEW

The SCR system is an exhaust gas aftertreatment solution used to reduce the nitrogen oxides within the exhaust gas.

For this purpose, a specified amount of DEF is injected into the exhaust system, dowstream of the DPF. The injected DEF into the exhaust system is converted to ammonia (NH_3) and carbon dioxide (CO_2). The resulting ammonia(NH_3) is used within a special catalyst in the exhaust stream. The resulting reaction converts the unwanted nitrogen oxides (NO_x) into harmless nitrogen (N_2) and water (H_2O) vapor.

DESCRIPTION

SELECTIVE CATALYST REDUCTION (SCR) SYSTEM SCHEMATIC DIAGRAM



ITEM	DESCRIPTION
1	Engine Control Module (ECM)
2	DEF tank with DEF tank module
3	DEF injection pump
4	DEF pressure line
5	Post - SCR NOx sensor
6	Post - SCR soot sensor (NAS market vehicles only)
7	SCR catalytic converter
8	Mixer plate
9	DEF injector
10	Pre - SCR NOx sensor (NAS market vehicles only)
11	DPF
12	Catalytic converter
13	Engine

The SCR catalytic converter reaches the operating temperature at 150 °C

(302°F). The temperature of the exhaust gas is measured by the post DPF exhaust gas temperature sensor, which is connected to the ECM.

The DEF injection pump supplies the DEF from the DEF tank at a pressure of 6.5 bar to the DEF injector via the heated DEF pressure line. The DEF injector is controlled with a Pulse Width Modulation (PWM) signal by the ECM.

The injected DEF is carried along by the exhaust gas flow and is evenly distributed in the exhaust gas by the mixer plate. The mixer plate is located in the exhaust pipe upstream of the SCR catalytic converter and downstream of the DEF injector. After the injection, the DEF is converted into ammonia (NH_3) and carbon dioxide (CO_2) during chemical reactions.

In the SCR catalytic converter, the ammonia (NH_3) reacts with the nitrogen oxides (NO_x) to produce nitrogen (N_2) and water (H_2O) vapor. The SCR system's efficiency is registered by the post SCR NO_x sensor.

In order to optimise the SCR system’s efficiency, the correct amount of ammonia (NH_3) is required in the exhaust gas. The ECM achieves this by operating in two modes:

- Storage mode
- On-line mode

For example, storage mode will operate during low speed driving condition; the on-line mode will operate at high speed driving condition.

Storage mode

When the system operates in storage mode, the ammonia (NH_3) is stored on the SCR catalytic converter and used as a function of the NO_x feed. In this mode, the objective of the control system is to ensure that a pre-determined amount of NH_3 is available on the SCR catalytic converter at any given time. As a consequence, it may be difficult to diagnose dosing system problems when the system is in storage mode.

On-line mode

When the system operates in on-line mode, the NOx is measured by the pre-SCR NOx sensor (or the model value) and the amount of ammonia (NH3) is injected as a function of the NOx feed. In this mode, the dosing system is ‘easy’ to diagnose if it is functioning correctly as the pulses from the DEF injector are frequent.

COMPONENT DESCRIPTION

DIESEL EXHAUST FLUID (DEF)

Diesel Exhaust Fluid (DEF) is a pure, odorless, colorless, synthetically manufactured, 32.5% aqueous solution of urea, used for the aftertreatment of exhaust gases in an SCR catalytic converter.

The SCR catalytic converter can be contaminated by low quantities of metals and thus the quality of the DEF fluid is held to closely controlled standards. DEF cannot be substituted by urea used in agriculture or diluted with any other fluid.

DEF is not categorized as a dangerous substance, it is non-flammable and non-toxic, there is no danger in the event of spills. DEF can be stored on board vehicles, despite the limitation that it crystallizes at temperatures below -11°C (12°F). In Europe, DEF is also known as AdBlue®, the fluid is specified by ISO22241.

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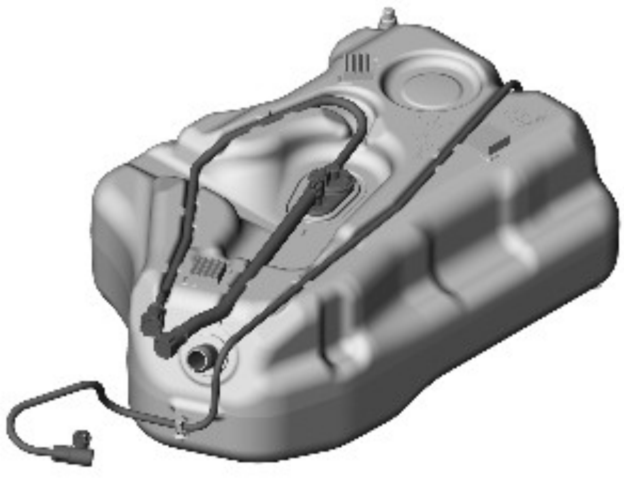
DEF TANK



E168828

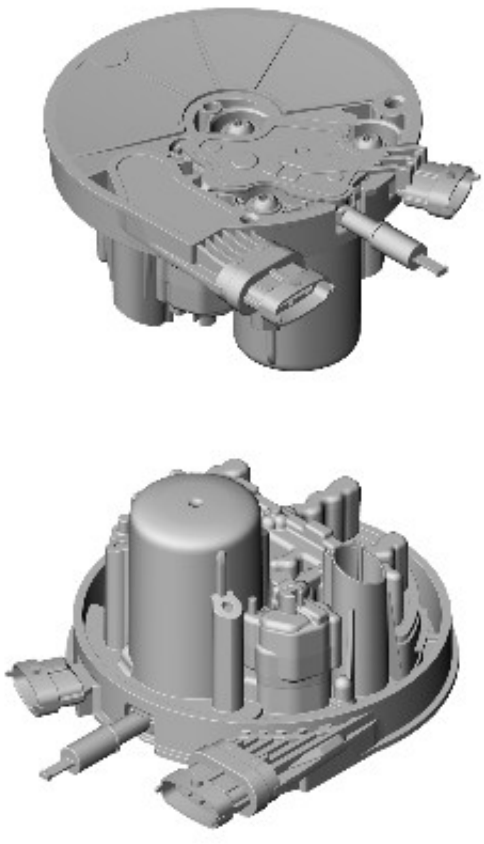
The DEF tank unit is blow moulded from High Density Polyethylene (HDPE). The tank is located on the left side of the vehicle, in front of the fuel tank, and mounted in a shield bolted to the underside of the vehicle floor pan. An additional shield covers the DEF supply module. The DEF tank unit contains the DEF supply module which is welded into the tank, and supplied as a unit. The tank includes a 12% unusable volume to protect the internal components as a result of fluid expansion under freezing conditions.

If the ambient temperature falls below -11°C (12°F), the DEF will freeze in the DEF tank, this will provide difficulty with the refill procedure. In order to thaw the DEF in the DEF tank, place the vehicle in a warm place for up to 2 hours before attempting to refill the tank.

The volume of the DEF tank is different on all JLR products, but has been designed so that refills are minimized outside of the vehicle service intervals.

When the vehicle consumes more DEF than anticipated, for example extended vehicle use in extreme temperatures, 'at altitude' or aggressive drive cycles, a warning message will be displayed in the Instrument Cluster (IC) message center to add DEF to the tank.

DEF TANK MODULE



E168829

ITEM	DESCRIPTION
1	Heater element
2	DEF level sensor
3	Electrical connection

The DEF tank module is located at the bottom of the DEF tank. The moduler is welded into the tank and can only be replaced as a complete assembly. The DEF tank module includes a life-time fit filter, a fluid level and temperature sensor and a heater element to defrost the DEF in extreme cold climates. The DEF level sensor is an ultrasonic device located in the DEF tank module, which sends the DEF level value to the ECM via PWM signals. The ultrasonic cone angle is 10° and the readings are most accurate when the vehicle is stationary and on level ground.

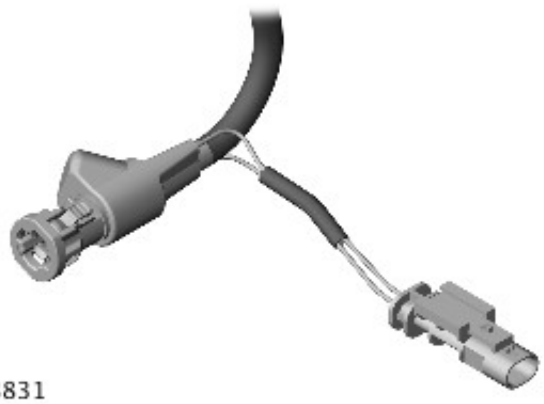
The heater element is a Positive Temperature Coefficient (PTC) type heater, which provides safe operation to the system. Increased heater element temperature results in decreased current drawn from the DEF heater control module, which actuates the power supply of the heater element. Under normal operation the maximum current is 6A.

DEF INJECTION PUMP

E168830

The DEF supply module is an assembly of two solenoid pumps, located in the DEF tank module. The pressure pump provides 6.5 bar operating pressure. The 6.5 bar pressure is required to maintain the complete DEF atomization in the exhaust gas. The purge pump is used to purge the DEF from the DEF pressure line at engine shut-down to prevent freezing in the DEF injector at low temperatures. The pumps have a fused power supply from the Engine Junction Box (EJB). The ECM controls the ground connection of the pumps individually via hardwired connections.

DEF PRESSURE LINE



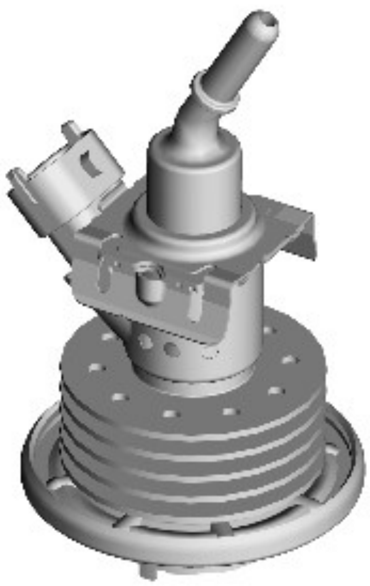
E168831

The DEF pressure line provides connection between the DEF injection pump and the DEF injector. The DEF pressure line is manufactured from a plastic material which is specifically designed for use with DEF.

A Copper Nickel Alloy resistor wire is installed within the DEF pressure line with an electrical connector. The resistor wire enables electrical heating of the DEF at low-ambient temperatures.

The DEF pressure line has hardwired connections to the DEF heater control unit, which actuates the power supply for the heater element, controlled by the ECM via the Private CAN bus.

DEF INJECTOR



E168832

The DEF injector is located in the exhaust system downstream of the DPF and it is secured to the S-shaped exhaust pipe with a clamp. Due to the position of the injector on the S-shaped exhaust pipe, the DEF is injected axially to the exhaust gas flow direction ensuring the DEF is mixed well and distributed evenly within the exhaust gas. The DEF injector consists of an injector and a passive cooling heat sink, this protect the injector from overheating due to the high exhaust temperatures.

The DEF injector works at high pressures to obtain the complete atomization of the injected DEF, this ensures the SCR catalytic converter is working to its optimum performance. The DEF injector is controlled by the ECM with PWM signals.

DEF HEATER CONTROL UNIT

E168833

The DEF heater control unit is located on the right side of the luggage compartment, behind the side trim panel. The heater system enables rapid SCR system operation in the event of frozen DEF and ensures an adequate quantity of defrosted DEF is available at all operating points. The DEF heater control unit has a switched power supply from the EJB, which is controlled by the ECM.

The DEF heater control unit has hardwired connections to the DEF tank module and the DEF pressure line heater element. If the ambient air temperature is below -5°C (23°F), the ECM switches on the DEF heater control unit via the Private CAN bus. The DEF heater control unit energizes the heater elements, located in the DEF tank module and in the DEF pressure line. The DEF heater control unit has on-board diagnostics to detect and report faults to the ECM via the Private CAN bus.

NITROGEN OXIDE (NOX) SENSOR

E168834

The Post SCR NOx sensor is located in the exhaust pipe downstream of the SCR catalytic converter. On NAS market vehicles there is an additional NOx sensor fitted upstream of the SCR catalytic converter. The NOx sensor comprises a sensor element attached to a dedicated control module with a hardwired connection.

The sensor and the control module are replaced as an assembly. The NOx sensor is an evolution of the wide band oxygen sensor. The sensor element is constructed from special ceramics and contains two oxygen density detecting chambers that work together to determine the NOx concentration in the exhaust gas. The control module sends a message via the Private CAN bus to the ECM for monitoring the effectiveness of the SCR System.

POST SCR SOOT SENSOR (NAS MARKET VEHICLES ONLY)

E168835

The post SCR soot sensor is located in the exhaust pipe downstream of the SCR catalytic converter. The sensor measures the soot particles in the exhaust gas to evaluate the DPF functionality.

The soot sensor comprises a sensor element attached to a dedicated control module with hardwired connection. The sensor and the control module can be replaced as an assembly. The adsorbed soot particles form conductive paths between the electrode combs in the sensor element. The control module measures the resistance between the electrode combs, and sends a message to the ECM via a Private CAN bus to evaluate the DPF functionality.

The control module regularly overheats the electrode combs to regenerate the sensor element.

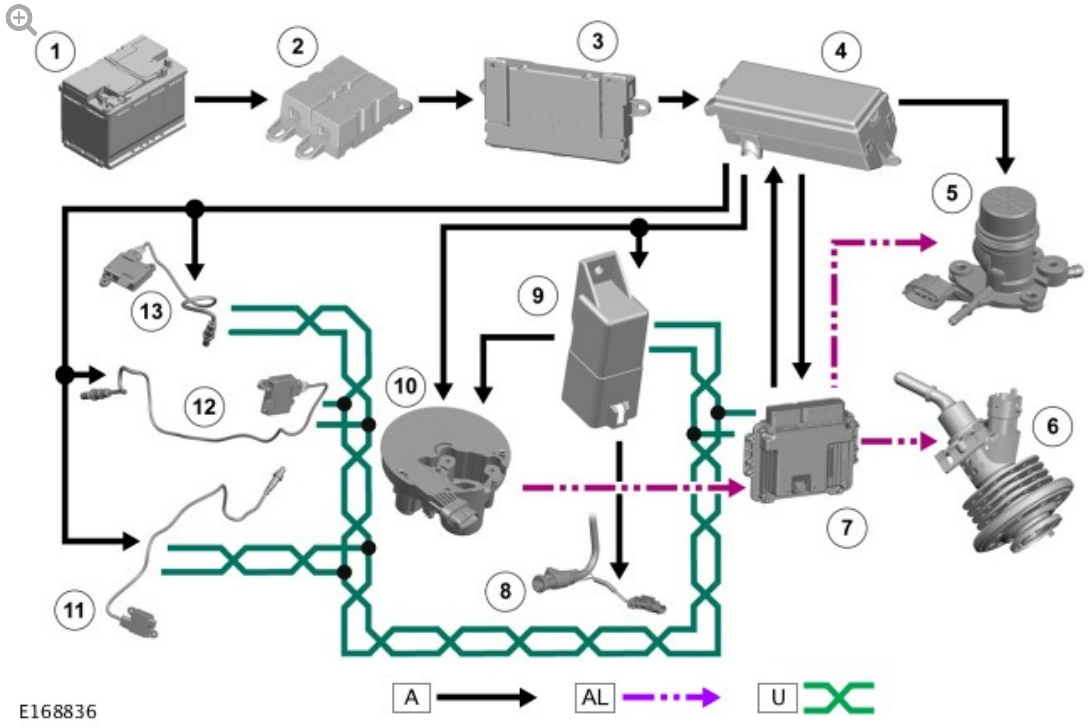
WARNING MESSAGES

It is a legal requirement for the driver to be informed of SCR system faults. If the performance of the SCR system is reduced a message will be displayed in the IC message center . There are three possible routes to receive a fault message on the IC:

- Low DEF volume in the tank
- Incorrect DEF quality supplied to the SCR System
- Efficiency of the SCR System is reduced in the event of a system fault.

In case of low DEF level a maximum distance of 800km (497 miles) is given to visit an authorised Land Rover dealer to refill the DEF tank. If the DEF fluid is not replenished within the 800km (497 miles) range and the engine is turned off the vehicle will fail to start. In this event customers can use two standard size refill bottles as a short term solution to restart the vehicle, a complete refill of the DEF tank will still be required by an authorised Land Rover dealer.

CONTROL DIAGRAM



A = HARDWIRED; AL = PULSE WIDTH MODULATION (PWM); U = PRIVATE CONTROLLER AREA NETWORK (CAN) BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Engine Junction Box (EJB)
5	DEF Pump Module
6	DEF Dosing Module
7	ECM
8	DEF pressure line
9	DEF Heater Control Unit
10	DEF Supply Module
11	Post SCR NOx sensor
12	Post SCR soot sensor (NAS market vehicles only)
13	Pre SCR NOx sensor (NAS market vehicles only)